Assessment - 1 of 3

PH6170 : Computational Physics

March 13, 2026

**Total Marks:** 100 **Duration:** 1.5 hrs

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# Q1 Generating and Verifying a Gaussian Distribution

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## Background

Random number generators in most programming environments typically produce numbers uniformly distributed in the interval [0,1]. In many scientific simulations, however, it is necessary to generate random numbers drawn from a Gaussian (normal) distribution.

One commonly used method for generating Gaussian random numbers from uniformly distributed random numbers is the Box–Muller transformation.

Given two independent random numbers $U_1$ and $U_2$ uniformly distributed in (0,1), the Box–Muller transformation produces two Gaussian random variables

$$Z_1 = \sqrt{-2\ln U_1}\cos(2\pi U_2)$$
$$Z_2 = \sqrt{-2\ln U_1}\sin(2\pi U_2)$$

which follow a normal distribution with zero mean and unit variance.

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## Tasks

1. Random number generation

   Generate 10,000 Gaussian random numbers using the Box–Muller transformation.

2. Histogram and probability density

   Construct a histogram of the generated random numbers using an appropriate number of bins.

   Convert the histogram into a probability density function (PDF).

3. Statistical properties

   Compute the following quantities from the generated dataset:

   - Mean
   - Variance

4. Verification

   Overlay the analytical Gaussian probability density function.

$$P(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

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**Food for thought**

- How closely does the generated distribution match the analytical Gaussian distribution?
- How does the agreement improve as the number of generated samples increases?

**P.S.** *These questions are intended to guide interpretation of the numerical results.*

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# Q2 Finite difference for numerical differentiation

## Background

Finite difference schemes are widely used to approximate derivatives in numerical simulations and data analysis. The accuracy of these schemes depends on the discretisation step size $h$. While decreasing $h$ generally reduces the truncation error of a finite difference approximation, very small step sizes may introduce numerical errors due to finite precision arithmetic. To explore these effects, consider the function

$$f(x) = e^{-x} cos(x)$$

## Tasks

1. Analytical derivative

   Determine the analytical expression for the first derivative $f'(x)$ at $x = 0.5$

2. Numerical Derivative

   Estimate the derivative (**only at** $x = 0.5$) using **central difference method** with **second and fourth order convergence** for the following step size.

   $$h = 10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}, 10^{-12}$$

3. Error estimation

   For each value of $h$, compute the absolute error (for $f'(x)$ estiamted at $x = 0.5$).

   $$E(h) = |f'_{\text{numerical}} - f'_{\text{exact}}|$$

4. Convergence Behaviour

   Plot **log (E) vs log(h)** and estimate the order of convergence from the slope of the curve in the region where the error decreases with $h$.

### Food for thought

- Does the error decrease indefinitely as $h$ becomes smaller?
- At approximately what step size does the error become minimal?

# Q3 Settling velocity of particles in viscous fluid

## Background

When small spherical particles settle through a viscous fluid, their terminal velocity depends on particle size and fluid properties. Experimental measurements often provide velocity values only for discrete particle sizes. The following measurements were recorded during a settling experiment.

| **Particle radius $r$ (mm)** | **Settling velocity $v$ (mm/s)** |
|---|---|
| 0.2 | 0.18 |
| 0.4 | 0.72 |
| 0.6 | 1.60 |
| 0.8 | 2.85 |
| 1.0 | 4.45 |

Table 1: Measured Data for Particle Settling

## Tasks

1. Using the interpolation routine implemented in your assignment, estimate the settling velocity at

$$r = 0.5 \text{ mm}$$

2. Construct the interpolation function $v(r)$ using all available data points.
3. Plot the interpolated velocity profile in the range

$$0.2 \leq r \leq 1.0$$

4. Using the same interpolation function, estimate the settling velocity at

$$r = 1.2 \text{ mm}$$

5. Plot the interpolated curve together with the measured data points.

**Food for thought**

- Why might the extrapolated velocity outside the measurement range be unreliable?
- Do all interpolation schemes behave similarly when extrapolated?

# Q4 Finding root in non-linear oscillatory system

## Background

Many non-linear equations encountered in physics cannot be solved analytically and must instead be solved using numerical root-finding techniques. Different numerical methods may exhibit markedly different convergence behaviour depending on the structure of the function and the choice of initial guess. Consider the non-linear equation:

$$f(x) = x^{\frac{1}{3}} - x$$

1 -2, -2

## Tasks

1. Plot the function $f(x)$ in the range

$$-2 \leq x \leq 2$$

   and identify the approximate locations of the roots.

2. Apply the **Newton–Raphson method** to determine a root of the equation starting from the initial guess

$$x_0 = \pm 0.01$$

   Continue the iterations until the absolute difference between successive iterates is less than $10^{-6}$.

3. Apply the **bisection method** to determine the root using the same tolerance.

4. Report the obtained roots and the number of iterations required by each method.

5. Plot the function together with the identified root(s).

### Food for thought

- How does the behaviour of the Newton–Raphson method compare with the bisection method for this function?
- What role does the derivative of the function play in determining the convergence of the Newton–Raphson method?

# Q5 Yet another interpolation problem

## Background

Experimental measurements sometimes provide only discrete samples of a function over a wide domain. Numerical interpolation can reconstruct the function between measurement points, but the behaviour of the interpolation may depend strongly on the numerical method employed.

| x | f(x) |
|---|---|
| -5 | 0.0016 |
| -4 | 0.0025 |
| -3 | 0.0044 |
| -2 | 0.0099 |
| -1 | 0.0385 |
| 0 | 1.0000 |
| 1 | 0.0385 |
| 2 | 0.0099 |
| 3 | 0.0044 |
| 4 | 0.0025 |
| 5 | 0.0016 |

## Tasks

1. Construct an interpolation function using **all available data points**.
2. Plot the interpolated function over the range

$$-5 \leq x \leq 5$$

   using at least **200 evaluation points**.
3. Estimate the value of the function at $x = 0.5$.
4. Plot the interpolated curve together with the data points.

**Food for thought**

- Does the interpolated curve behave smoothly across the entire interval?
- Are there regions where the interpolation deviates unexpectedly from the trend suggested by the data?

*Numerical methods do not merely approximate physics -*
*they shape the physics you believe you see.*